

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
8th Semester of B. Pharm. Examination
University Theory Examination October 2015
PH 413 Novel Drug Delivery System

Date: 20/10/2015 , Tuesday

Time: 1.30 p.m to 4.30 p.m

Maximum Marks: 80

Instructions:

- 1. Figures to right indicate marks.**
- 2. Section wise separate answer book to be used.**
- 3. Draw neat sketches wherever necessary.**

SECTION- I

Q 1 Attempt any TWO of the following

- A** Explain lag time, burst effect and reservoir systems w.r.t. controlled release formulations. Describe the ideal requirements for sustained release formulations. **10**
- B** Give the difference between conventional and controlled release system. What is the major objective of controlled drug delivery system? Give advantages and disadvantages of such a system. **10**
- C** Define Nanoparticles. Enumerate various methods for preparation of nanoparticles and describe solvent emulsification evaporation method for in detail. **10**

Q 2 Attempt any TWO of the following

- A** What properties are required for the drug to be a candidate for transdermal drug delivery system? Describe formulation and evaluation of transdermal drug delivery system in brief. **10**
- B** Define liposomes and niosomes. Describe their evaluation in length. **10**
- C** Discuss microparticles as a drug delivery system in detail. **10**

SECTION- II

Q 3 Attempt any TWO of the following

A Write a short note on:

1. ADEPT technology. **10**

2. Monoclonal Antibodies.

B 1. Explain the main concept of Drug targeting with an example. **10**

2. Differentiate Active targeting and Passive targeting.

C Write a short note on: Immediate release delivery systems. **10**

Q 4 Attempt any TWO of the following

A Write a note on:

1. Osmotic pressure controlled drug delivery system. **10**

2. Gastro retentive drug delivery system.

B Describe in brief the classification of oral controlled drug delivery system. **10**

C Write short note on:

1. Iontophoresis. **10**

2. Sonophoresis.

3. Micro needle array.
